A Typology of Lobbying Strategies

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Abstract

This memo presents a simplified typology of lobbying strategies organised around two primitives—alignment with the pivotal voter bloc and relative power—and three nested levels of contestation: problem structure, constraints, and policy choice. The typology is applied to climate politics, where it identifies four strategic regimes (Amplify, Activate, Steer, Reframe) and traces the population shift across them as the energy transition proceeds. Four macro-institutional features—interest intermediation, electoral systems, veto points, and bureaucratic autonomy—are shown to condition how the primitives translate into observable strategies. The memo concludes by demonstrating that the typology has direct implications for the credible estimation of lobbying effects: it specifies the confounding structure, predicts the direction and source of bias in existing estimates, identifies the dependent variables on which effects should register at each level, and motivates a partial-identification strategy using Manski bounds tightened by theory-derived monotonicity assumptions.

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1 The Framework

1.1 The problem

An interest group seeks to shift policy. The decision-maker—politician or administrator—is constrained by voter preferences, by other organised groups, by institutional rules, and by state capacity. Lobbying is the activity of operating on the decision-maker, on the constraints, or on the structure of the problem itself. The group's strategic choice depends on two variables: its *alignment* with the electorally pivotal voter bloc, and its *power* relative to opposing organised interests.

1.2 Three levels of contestation

All lobbying strategies sit on three nested levels. Each contains those below it as a special case; each is more consequential but harder to observe.

Level 3: Policy choice. Taking both the salient dimension and the constraints as given, the actor shifts the decision-maker's choice through exchange, information, or political pressure. This is the most visible form of lobbying and the one most commonly measured: regulatory comments, political donations, legislative testimony. Its scope is limited to the parameters of a policy whose existence and framing are already determined.

Level 2: Constraint configuration. Taking the salient dimension as given, the actor reshapes the constraints that discipline the decision-maker. Three sub-types:

- *Preferences*: shifting what voters want on the operative dimension (information campaigns, narrative construction, framing).
- *Attention*: shifting how much voter preferences count electorally (politicisation, de-politicisation, issue activation, issue suppression).
- *Capacity*: shifting what the state can implement and how independently it can do so (budget lobbying, litigation, procedural obstruction, informational substitution, capacity-building).

Level 1: Problem structure. The actor shifts which dimension of the policy space is politically salient—what the conflict is *about*. This is distinct from preference shaping because it changes *which* preferences are operative rather than shifting preferences within a fixed frame. Coalitions that are allied on one dimension may be divided on another; an actor that succeeds at Level 1 can therefore restructure the entire political landscape.

Epistemic note. Level 1 is the typology's most theoretically distinctive claim but its most empirically unvalidated. Existing evidence is consistent with the logic but does not directly test whether actors *strategically choose* which dimension to contest. The causal arrow in the available studies runs from policy instrument to mobilisation pattern, not from lobbying strategy to salient dimension. A simpler explanation—that firms respond mechanically to what directly affects them—may account for much of the observed variation without requiring the Level 1 apparatus.[1]

1.3 Two arenas

The political arena. The decision-maker aggregates voter and interest-group preferences under electoral visibility. In parliamentary systems, this operates in two stages: intra-party (groups shape party positions through policy committees and advisory bodies) and inter-party (electoral competition aggregates party positions into government policy).

[1] Møller (2025) finds that green industrial policy mobilises low-carbon sectors while traditional regulation mobilises high-carbon sectors. This is consistent with the Level 1 logic but tests the downstream consequence (mobilisation conditional on instrument), not the upstream strategy (lobbying over instrument choice).

The political arena favours aligned actors, public-facing influence capacity, and issues where electoral salience benefits the actor.

The administrative arena. An administrator chooses implementation intensity $\alpha \in [0, \bar{\alpha}(C, \iota)]$, where C denotes *capacity quantity* (staff, budgets, analytical resources) and $\iota \in [0, 1]$ denotes *capacity independence*—the share of the regulator's information base that is independent of regulated entities. The feasibility frontier $\bar{\alpha}$ is increasing in both arguments: a regulator with high C but low ι is constrained not by what it can do but by what it can evaluate independently.

This two-dimensional representation addresses an important empirical finding: a regulator with extensive technical information (C high) but near-total informational dependence on incumbents (ι low) may be more captured than a regulator with lower but independent capacity.[2] Four capacity-shaping mechanisms are distinguished:

- *Budget erosion*: reducing C by cutting regulatory budgets or staffing.
- *Procedural obstruction*: reducing effective C through litigation and compliance burdens.
- *Informational substitution*: reducing ι by becoming the dominant source of technical information on which the regulator depends. Capacity *provision* that creates informational dependence.
- *Capacity building*: increasing both C and ι by helping construct genuinely independent regulatory infrastructure. Rare, but predicted when the actor benefits from effective, credible regulation.

The administrative arena favours misaligned actors seeking to avoid public exposure, actors with technical expertise, actors with strong relational networks, and issues where implementation details determine effective policy stringency. A critical measurement limitation: influence in the administrative arena operates through both *formal* channels (consultation responses, regulatory comments) and *relational* channels (personal networks, revolving-door access, shared professional cultures). The relational channel is empirically consequential but structurally unobservable through standard proxies.

1.4 Two determinants and four regimes

Alignment (α). The distance between the group's preferred policy outcome and the pivotal voter bloc's bliss point on the salient dimension. Observable through material-interest indicators: emissions intensity, supply-chain carbon dependence, portfolio composition, cost-incidence exposure. The alignment variable is the framework's best-supported predictor: across the studies reviewed below, material interests predict lobbying strategy more consistently than any other variable.

[2] Lockwood, Mitchell, and Hoggett (2020) document this pattern in the GB Capacity Market, where DECC officials relied heavily on incumbent-provided data and analysis for the design of demand-side response rules.

Power (ρ). The group's capacity to affect outcomes relative to opposing organised interests, encompassing both instrumental resources (lobbying expenditure, technical staff, media capacity) and structural weight (employment, investment, exit credibility). A systematic limitation: the relational component of power (personal networks, cultural alignment with regulators) is empirically consequential but unobservable through standard proxies.

These two variables define four strategic regimes:

Table 1: Four strategic regimes.

Regime	Alignment	Power	Predicted strategy
Amplify	High	High	Level 3 (contest policy directly). Political arena. Public channel. Broad coalition. The actor benefits from salience and converts existing support into policy.
Activate	High	Low	Level 2 attention (raise electoral salience). Political arena. Public channel. Broad coalition. The actor's allies exist but are politically dormant.
Steer	Low	High	Level 1 or Level 2 (reshape problem or constraints). Administrative arena. Direct channel. Narrow coalition. The actor cannot win publicly but need not.
Reframe	Low	Low	Level 1 (shift to a dimension where the actor is better positioned), if feasible. If not viable: *strategic exclusion.*

The strategic viability filter. Before entering the decision logic, an actor must possess sufficient influence capacity on at least one channel to have a non-negligible probability of affecting outcomes. When an actor is below threshold on all channels—weak direct access, weak public credibility, weak organisational infrastructure—the framework predicts no viable strategy. This is not a trivial edge case: for small, fragmented entrants facing concentrated incumbents in low-salience administrative arenas, strategic exclusion may be the modal outcome.

1.5 Dynamics

Let $D_t = (\alpha_t, \rho_t)$ be the vector of primitives, p_t the policy outcome, and m_t the state of autonomous market and technological change. The expanded feedback operator maps:

$$D_{t+1} = F(D_t,\, p_t,\, m_t). \tag{1}$$

Policy outcomes shift alignment and power through four channels: (i) the *alignment channel*, through which policy and market dynamics change bliss points by altering material interests, cascading through supply chains with instrument-specific speed; (ii) the *power channel*, through which policy outcomes alter the political weight of organised interests; (iii) the *capacity channel*, through which policy outcomes change administrative capacity and independence; and (iv) the *credibility channel*, through which repeated self-interested messaging erodes an actor's public persuasion credibility, pushing it towards direct or mobilisation strategies.

The inclusion of m_t acknowledges that bliss points are shifted by autonomous technological cost declines, competitive market dynamics, and strategic firm investment—not only by policy feedback.[3] The static typology is the special case $F = \text{id}$ and $m_t = \text{const}$.

1.6 Empirical status

The framework's components vary substantially in their empirical grounding:

- **Well supported:** the alignment variable as the primary predictor of strategy; the two-arena distinction; the feedback logic (policy and market dynamics shift the primitives across rounds).
- **Indirectly supported:** Level 1 dimension selection as actor strategy (vs. mechanical response to policy); quiet politics as strategic counter-mobilisation avoidance (vs. institutional default).
- **Conjectured:** the persuasion/mobilisation sub-distinction within the public channel; Bayesian persuasion constraints; counter-mobilisation dynamics; mobilisation as pivotal-bloc restructuring; credibility decay driving within-channel adaptation.

The conjectured elements represent the framework's research frontier. They are presented as testable predictions, not as established theoretical claims.

[3] Kim (2022) documents that firms' carbon-reduction investments (chosen for commercial reasons) shift their bliss points and hence their lobbying strategy independently of policy.

2 Application: Climate Lobbying

2.1 Structural features

Climate policy exhibits three structural features that condition the strategic landscape.

Pivotality–alignment mismatch. The voter bloc most often pivotal in climate policy—cost-sensitive swing voters concerned with energy prices, household costs, and employment—is structurally misaligned with the pro-stringency coalition. Climate-concerned voters are numerous but electorally diffuse. This mismatch means that pro-climate actors face an inherent disadvantage in the political arena unless they can either persuade the cost-sensitive bloc (Level 2 preference shaping) or activate the climate-concerned bloc sufficiently to displace it from pivotality (Level 2 attention shaping).

Two-dimensionality. Climate policy varies along at least two dimensions: *stringency* (how much emissions reduction) and *incidence* (who bears the cost). Coalitions stable on one dimension may fracture on the other. Utilities that support carbon pricing (stringency) may oppose specific allocation rules (incidence). Energy-intensive industry that accepts stringency in principle contests free-allocation formulae, border-adjustment mechanisms, and competitiveness safeguards. This two-dimensionality makes Level 1 strategies especially consequential: an actor that shifts salience from stringency to incidence can split the pro-climate coalition.

Supply-chain propagation. Carbon-intensive supply chains create an extended anti-stringency coalition beyond direct emitters. Cory, Lerner, and Osgood (2021) find that 66% of US climate lobbying originates from firms outside the highest-emitting industries, connected through supply-chain linkages. The effective bliss point of a downstream firm reflects the carbon intensity of its upstream partners, broadening the anti-stringency coalition well beyond the firms that directly emit.

2.2 The four regimes in climate politics

2.2.1 Amplify: mature renewables and the clean-economy coalition

The clearest current occupant is the mature renewables industry in jurisdictions where clean energy is cost-competitive. Its preferred policy (continued clean-energy support, rising carbon prices) aligns with what the cost-sensitive pivotal bloc can accept, because renewables are now the cheap option in many markets. The industry has growing political weight: employment, investment, and supply chains in electorally relevant districts.

Strategy: Level 3, political arena, broad coalition, public channel. These actors benefit from salience and want climate policy to be an electoral issue because they win that

contest. The US Inflation Reduction Act coalition—clean-energy firms, manufacturing unions, and politicians from districts with clean-energy investment—is the paradigmatic case.

This regime is historically new. For most of the history of climate politics, no major economic actor occupied Amplify. The regime became populated only as the clean economy grew large enough to generate independent political weight. Møller's (2025) finding that green industrial policy mobilises the low-carbon sector is evidence of this regime filling.

2.2.2 Activate: environmental movements and early-stage clean technology

Environmental NGOs have occupied this regime for decades. Their preferences align with the climate-concerned public, but that public is politically dormant relative to its size: people express concern but do not vote on the issue. NGOs lack concentrated economic power.

Strategy: Level 2 attention shaping, political arena, broad coalition. The objective is to raise the electoral salience of climate so that politicians treat it as a vote-winning issue. Observable as: protest movements (Fridays for Future, Extinction Rebellion), voter registration drives, electoral scorecards, and reframing climate as a household issue (health, extreme weather, insurance costs).

Early-stage clean-technology firms also occupy this regime before they acquire political weight. They are aligned but individually too small to matter; they depend on activation of the broader clean-economy constituency.

The boundary between Activate and Amplify is blurring in real time: actors that were in Activate five years ago are moving to Amplify as the clean economy grows. This is the feedback operator at work.

2.2.3 Steer: fossil fuels, energy-intensive industry, and quiet opposition

This regime has dominated climate politics historically. Two sub-populations with different strategies occupy it.

Fossil-fuel incumbents. Deeply misaligned on stringency, very strong. Their trajectory within the Steer regime illustrates the credibility channel of the feedback operator:

- *1990s–2000s*: Level 2 preference shaping via public persuasion (climate denial). An unusually public strategy for a misaligned actor, sustainable only because of overwhelming resource superiority. As scientific consensus hardened, credibility collapsed.
- *2000s–2010s*: Shift to Level 1 dimension selection. Cease contesting whether climate change is real; begin contesting which instrument is appropriate. Support carbon pricing (demand-side, with incidence falling on consumers and stringency dilutable

administratively) and oppose supply-side regulation (which directly threatens asset values).

- *2010s–present*: Increasingly administrative. Quiet lobbying on implementation details, allocation rules, compliance flexibility, phase-in timelines. Informational substitution: providing the technical analysis that regulators lack independent capacity to generate.

Throughout, the coalition form is narrow (individual firm or sector association) and the channel is direct.

Energy-intensive industry. A different variant: these actors accept stringency in principle but contest incidence. The relocation threat functions as attention shaping on the competitiveness dimension—activating cost-sensitive voters who already worry about jobs. Strategy: mobilisation on the competitiveness frame combined with direct engagement on allocation design.

Quiet opposition coalitions. In corporatist systems, business and labour jointly exercise insider influence to keep climate policy low-salience. This is explicitly anti-attention: preventing the issue from becoming an electoral contest, because if it does, the misaligned coalition loses. The strategy is strategic, not merely institutional—although the institutional context (corporatism) makes it cheap.

2.2.4 Reframe or exit: entrants, communities, and the strategically excluded

DSR providers. Lockwood et al.'s (2020) demand-side response providers exemplify strategic exclusion. Misaligned with incumbents (they sought market access that incumbents sought to deny), weak on all channels (small, fragmented, no relational networks with regulators, no public-channel capacity), they had no viable strategic path. The Capacity Market rules were written without effective input from the actors most affected by them.

Coal-dependent communities. Misaligned with the direction of policy and weakening as the political coalition defending coal shrinks. The "just transition" framing is a Level 1 strategy: shift salience from stringency to distributional fairness. When reframing fails, exit is economic (community decline, outmigration) rather than political.

Small island developing states. Deeply aligned with stringency but weak relative to the major emitters that control the international process. Their strategy has been Level 1 reframing: shift the debate from "economic costs of mitigation" to "existential threat and loss and damage." Partially successful—loss and damage is now on the international agenda—but the power asymmetry constrains what reframing alone can achieve.

Labour in carbon-intensive industries. Individually weak, but capable of regime upgrading through coalition. Mildenberger's (2020) "double representation"—labour unions amplifying fossil-fuel interests through a second political channel—moves workers from Reframe to Steer by allying with capital that shares their incidence concern.

2.3 The dynamic story

The most important feature of climate politics when viewed through the four regimes is the *population shift across regimes over time*. For decades, the landscape was:

- Steer: densely populated (fossil fuels, utilities, heavy industry, allied labour).
- Activate: sparsely populated (environmental NGOs, climate scientists).
- Amplify: empty.
- Reframe: small actors on both sides.

The energy transition is repopulating the map. Amplify is filling (renewables, electric vehicles, clean-technology supply chains, green finance). Activate is growing (the climate-concerned public becoming more electorally active). Steer is thinning (some utilities and automotive firms migrating to Amplify as portfolios shift). Reframe is growing on the incumbent side (coal, some oil, fossil-dependent communities).

The feedback operator drives this: green industrial policy creates Amplify constituencies; carbon pricing restructures cost incidence and hence alignment; technological cost declines make renewables commercially dominant; each round of policy shifts the primitives for the next round. The political tipping point—the moment when the Amplify regime becomes more politically powerful than the Steer regime—may have been reached in the EU and is approaching in some US contexts, though the picture varies enormously by jurisdiction.

3 Institutional Scope Conditions

The two primitives—alignment and power—predict which regime an actor occupies. The institutional context predicts the *specific strategy* within that regime. Four macro-institutional features do most of the conditioning work.

3.1 Interest intermediation: pluralism versus corporatism

In corporatist systems, institutionalised insider access is the default channel. Peak associations have formal roles in policy development; the direct channel is structurally cheap. In pluralist systems, access is competitive and purchased through lobbying expenditure, campaign finance, and relational investment; the direct channel is structurally expensive.

Interaction with alignment. Corporatism pulls even *aligned* actors towards the direct channel, because institutional insider access is cheap relative to public mobilisation. The typology predicts that aligned actors should go public (the Amplify regime), but corporatism dampens this. The observable implication is that all four regimes are compressed towards the direct channel in corporatist contexts: public mobilisation is rarer across the board, not only for Steer actors. The framework's predictions about channel choice are more directly testable in pluralist systems.

Interaction with power. Corporatism compresses power differentials *within* the institutional arena (peak associations give small firms a collective voice) but amplifies the exclusion of actors *outside* the corporatist structure (new entrants, social movements, firms in emerging sectors without established associations). The viability filter bites differently: in pluralist systems, resources are the gatekeeper; in corporatist systems, institutional membership is. The *type* of double disadvantage differs accordingly.

3.2 Electoral systems: majoritarian versus proportional

In majoritarian systems, the pivotal bloc is geographically concentrated in marginal constituencies. An actor's alignment matters only if it is aligned with voters in *pivotal districts*, creating a geographic specificity premium: actors with geographically concentrated economic footprints (a factory in a swing district) have disproportionate influence relative to their aggregate economic weight. In proportional systems, the pivotal bloc is defined by coalition arithmetic, and there is no geographic specificity premium; aggregate national alignment is more directly operative.

Interaction with alignment. In majoritarian contexts, an actor can be nationally misaligned but locally aligned in pivotal districts. Energy-intensive industry in the US Rust

Belt is nationally misaligned with climate ambition but locally aligned with pivotal voters in Pennsylvania, Michigan, and Wisconsin. The alignment variable needs a geographic modifier: what matters is $|g_k - b_{j^*}|$ *in pivotal districts*, not in the aggregate. Sick's (2023) finding that MEP nationality predicts alignment with the automotive industry better than party affiliation illustrates the same logic in the EU context.

Interaction with power. Majoritarian systems amplify the power of geographically concentrated interests and dilute the power of geographically diffuse interests. This explains a puzzle the typology does not otherwise address: why the US has been slower on climate than Europe despite comparable clean-energy resources. The majoritarian system amplifies fossil-fuel interests concentrated in pivotal states and dilutes diffuse clean-energy interests. Under proportional representation, the same balance of interests would produce more pro-climate policy because the geographic concentration premium disappears.

3.3 Veto points

The number and configuration of veto points determines how many arenas exist and how actors exploit disagreement across them. In high-veto-point systems (the US, with presidentialism, bicameralism, judicial review, and federalism), misaligned actors benefit disproportionately: they need to win in only *one* veto point to block, while aligned actors must win across *all* veto points to enact. In low-veto-point systems (Westminster systems with parliamentary sovereignty), losing in parliament means losing altogether, with limited recourse.

Interaction with alignment. Veto points make the alignment variable *less* predictive: even a well-aligned, strong actor may fail to enact policy because opponents can find one veto point where the local configuration favours them. This is the central institutional insight of Mildenberger's (2020) analysis of US climate politics: the institutional structure systematically favours blocking coalitions.

Interaction with power. Veto points multiply the *uses* of power. A strong actor in a high-veto-point system can deploy resources across multiple arenas simultaneously (lobby Congress, shape agency rules, litigate in federal courts, work state legislatures), whereas a weak actor cannot. The power differential between strong and weak actors is therefore *amplified* by the number of veto points.

3.4 Bureaucratic autonomy

In systems with high bureaucratic autonomy (Scandinavian countries, the EU Commission in some domains), administrative agencies have independent technical capacity, professional

norms that resist political interference, and institutional protections. The variable ι is structurally high. In systems with low autonomy (the US in many regulatory domains, many developing countries), ι is structurally low—not because of informational substitution by regulated entities, but because of political control over the bureaucracy.

Interaction with alignment. High autonomy insulates the administrative arena from political alignment: a well-aligned actor that wins politically may find the bureaucracy implementing policy more rigorously than the political compromise intended. A misaligned actor that loses politically may find faithful implementation regardless of lobbying. Low autonomy collapses the two-arena distinction: the administrative arena becomes a subsidiary of the political arena, and alignment with the political principal translates directly into administrative influence.

Interaction with power. High autonomy raises the minimum capacity threshold for effective administrative lobbying. Informational substitution is harder when the regulator has independent analytical capacity. Conversely, low autonomy allows political power to convert directly into administrative influence without requiring technical expertise.

3.5 Summary

The primitives predict the regime; the institutions predict the specific strategy within the regime. Corporatism compresses all regimes towards the direct channel. Majoritarian elections make alignment district-specific. Veto points multiply arenas and systematically advantage blocking coalitions. Bureaucratic autonomy conditions the administrative arena's independence. A complete empirical test of the framework requires institutional context as a moderating variable, not merely as background.

4 The Inferential Problem

The typology has direct implications for the credible estimation of lobbying effects. This section argues that most existing empirical strategies produce biased estimates in ways that are *predictable from the framework itself*; that the bias structure implies specific dependent variables on which effects should register at each level; and that a partial-identification strategy using Manski bounds, tightened by theory-derived monotonicity assumptions, offers a more honest path forward than conventional point estimation.

4.1 The causal graph

The quantity of interest is the causal effect of lobbying by actor k on policy outcome Y:

$$\tau_k = Y(\text{lobbying} = 1) - Y(\text{lobbying} = 0). \tag{2}$$

The typology implies the following confounding structure. Let G denote the actor's preferences (determined by material interests), A alignment, P relative power, L lobbying activity (the treatment), and Y the policy outcome. The causal relationships are:

$$\begin{aligned}
&G \to A, \\
&A \to L, \quad A \to Y, \\
&P \to L, \quad P \to Y, \\
&L \to Y.
\end{aligned}$$

Alignment and power are common causes of both L and Y: aligned actors receive favourable outcomes even without lobbying (because they share preferences with the pivotal bloc), and powerful actors affect outcomes through structural power (investment decisions, employment, exit threats) independently of active lobbying. Any estimate of $L \to Y$ that fails to block both the $A \to Y$ and $P \to Y$ backdoor paths is biased.

4.2 Five sources of bias

4.2.1 Alignment confounding

Firms that lobby *for* climate regulation tend to be aligned with the pro-climate bloc. But aligned firms receive favourable policy outcomes partly because their preferences coincide with the pivotal bloc's—the policy would have moved in their direction absent lobbying. A regression of Y on L, even controlling for observable firm characteristics, attributes to lobbying what is partly the direct effect of alignment.

Direction: upward bias for aligned actors. Symmetrically, firms that lobby *against* climate regulation tend to be misaligned, and misaligned firms face unfavourable outcomes

because the pivotal bloc does not share their preferences. The estimated effect of anti-climate lobbying understates its actual effect (or estimates the wrong sign) because the alignment confound pushes outcomes in the opposite direction.

This produces a perverse artefact: studies tend to find that pro-climate lobbying "works" and anti-climate lobbying "does not," which may reflect confounding rather than differential effectiveness.

4.2.2 Structural power misattribution

Large, economically important firms lobby more (they have higher influence capacity) and also affect policy through structural power—the implicit threat of disinvestment, employment loss, and capital flight—independently of active lobbying. Most designs measure observable lobbying L and attribute all associated policy influence to it, even though much of the influence operates through a structural-power channel that lobbying proxies but does not constitute.

Direction: upward. Formally, the true model is $Y = \beta_1 L + \beta_2 P + \varepsilon$, but most studies estimate $Y = \gamma L + u$, where:

$$\gamma = \beta_1 + \beta_2 \cdot \frac{\mathrm{Cov}(L, P)}{\mathrm{Var}(L)}. \tag{3}$$

Since $\mathrm{Cov}(L, P) > 0$, $\gamma > \beta_1$: the estimated lobbying effect absorbs the structural-power effect.

4.2.3 Level mismatch

Most empirical studies measure Level 3 lobbying (expenditure, consultation responses, donations). But the typology predicts that the most consequential lobbying often occurs at Level 1 (dimension selection) and Level 2 (constraint shaping), which are less visible and harder to measure. This creates *attenuation bias*: if the measured variable (Level 3) is only weakly correlated with the theoretically consequential variable (Level 1 or Level 2), the estimated effect is attenuated towards zero. Studies that find "lobbying does not matter much" may be measuring the wrong level.

It also creates *mismatch bias*: if the dependent variable is a Level 3 outcome (a specific regulatory parameter) but the lobbying that mattered operated at Level 1 (which issues reached the agenda) or Level 2 (whether the issue achieved electoral salience), the study estimates the effect on the wrong outcome.

Direction: attenuation towards zero when measured lobbying is at the wrong level.

4.2.4 Arena selection

The typology predicts that actors select into arenas based on their primitives. Misaligned-strong actors select into the administrative arena; aligned actors select into the political arena. A study estimating lobbying effects in one arena observes a non-random sample selected on variables that also determine outcomes. If the study finds that administrative lobbying is associated with less stringent regulation, this may reflect the fact that misaligned actors select into the administrative arena *when political conditions allow them to extract concessions*, not that administrative lobbying is intrinsically more effective.

Direction: context-dependent, but generally producing an illusion of differential arena effectiveness that reflects selection rather than causal differences.

4.2.5 Observability bias

The relational component of administrative influence (personal networks, revolving doors) is empirically consequential but structurally unobservable. The measured variable L is a noisy proxy for total influence, and the noise is non-random: it systematically excludes the relational channel, which the typology predicts is most important in the administrative arena for strong actors.

Direction: upward bias on the estimated effect of formal lobbying (because formal lobbying proxies for relational influence), combined with systematic underestimation of total lobbying influence.

4.3 Over-estimation

Beyond the five bias sources identified above, lobbying effects are systematically over-estimated through several additional mechanisms.

Reverse causality from anticipated outcomes. Firms lobby on issues where they expect to have influence—because the political environment is favourable or because the policy is already moving in their direction. Lobbying is an endogenous response to anticipated success, not only a cause of it. Formally, if actors lobby when they anticipate Y will be favourable regardless, then $\mathbb{E}[Y \mid L = 1] > \mathbb{E}[Y \mid \text{do}(L = 1)]$: the observed association overstates the causal effect.

Sample selection on the dependent variable. Studies of lobbying effects examine issues where lobbying occurred and a policy outcome is observed. But the most effective lobbying may keep issues off the agenda entirely (Level 1 and Level 2 quieting). These cases never enter the sample. The studied cases are disproportionately those where Level 1 and Level 2 lobbying failed or was not attempted—biasing the sample towards Level 3 lobbying, which the typology predicts is the least consequential.

Credit-claiming. Some lobbying is strategically targeted at outcomes that will occur regardless, so that the lobbyist can claim credit to clients or members. The lobbying is real and observable; the causal effect is zero. A study measuring the association between lobbying and outcomes would code this as a success.

Case-study bias. Qualitative studies trace temporal sequences: firms lobbied for X, and X occurred. The inference is that lobbying caused X. But the typology implies that firms lobby for things that are already likely (alignment confounding) or that they would obtain through structural power alone. The case-study method lacks the counterfactual and therefore systematically favours narratives in which lobbying is consequential.

General principle. Lobbying effects are over-estimated whenever the lobbying decision is endogenous to the expected outcome. Since the typology implies that lobbying is *always* endogenous to the expected outcome (actors choose strategies based on the primitives, which also determine outcomes), the over-estimation problem is structural, not incidental.

4.4 Where effects should show up

The typology predicts that lobbying effects register on different dependent variables at each level. The correct DV depends on the level at which the measured lobbying operates. Level–DV mismatch produces null results that reflect measurement error, not genuine ineffectiveness.

Level 3 → policy design variables. When lobbying operates at Level 3, effects should register on specific design parameters: allocation formulae, exemption thresholds, phase-in timelines, compliance flexibility, monitoring protocols. These are the parameters the decision-maker adjusts within the constraints set by Levels 1 and 2. DV candidates: free-allocation shares in emissions trading, exemption coverage in environmental regulation, baseline methodologies in carbon-accounting rules.

Level 2 → constraint variables. When lobbying operates at Level 2, effects should register on variables measuring the constraints on policy choice:

- *Preferences*: public opinion on climate policy, media framing of climate issues, voter issue-salience rankings.
- *Attention*: electoral salience of climate, legislative-agenda composition, committee hearing frequency, party-manifesto content.
- *Capacity*: regulatory staffing and budgets, enforcement rates, ratio of independent to industry-sourced analysis, regulatory-fragmentation indices.

Level 1 → problem-structure variables. When lobbying operates at Level 1, effects should register on what dimension of the policy problem is being contested. DV candidates: which instruments are on the legislative agenda (carbon pricing vs. green industrial policy vs. supply-side regulation), how the issue is framed in elite discourse (cost vs. opportunity vs. security), which committee has jurisdiction.

The systematic pessimism bias. Studies that find "lobbying does not matter" may be matching the wrong level of measured lobbying to the wrong dependent variable. The typology predicts that effects are detected when the level of measured lobbying matches the level of the DV, and are attenuated or null when they do not.

4.5 Partial identification: Manski bounds

Given the severity of the confounding problem, point estimation of τ_k requires identifying assumptions that the typology tells us are violated. A more honest approach is partial identification: given the data and theory-derived assumptions, what is the *range* of causal effects consistent with the evidence?

Setup. Let $Y(1)$ be the policy outcome when actor k lobbies and $Y(0)$ the outcome when it does not. The average treatment effect is $\text{ATE} = \mathbb{E}[Y(1) - Y(0)]$. We observe $Y(1)$ for lobbyists and $Y(0)$ for non-lobbyists, but not the reverse. Assuming $Y \in [y_0, y_1]$, the worst-case Manski bounds are:

$$\begin{aligned} \mathbb{E}[Y \mid L{=}1] - \mathbb{E}[Y \mid L{=}0] - (y_1 - y_0)\, P(L{=}0) \;&\leq\; \text{ATE} \\ &\leq\; \mathbb{E}[Y \mid L{=}1] - \mathbb{E}[Y \mid L{=}0] + (y_1 - y_0)\, P(L{=}1). \end{aligned} \tag{4}$$

These are typically too wide to be informative. The typology provides structure for tightening them.

Monotone treatment response (MTR). The typology implies that lobbying weakly moves policy in the direction the lobbyist prefers: $Y_i(1) \leq Y_i(0)$ for anti-stringency lobbying (where Y is stringency). This assumption is credible for Level 3 direct lobbying in the administrative arena, where the mechanism is information provision and rule-shaping: a regulatory comment requesting a weaker standard cannot make the standard *stronger*. Under MTR, the upper bound on the ATE is the naïve difference $\mathbb{E}[Y \mid L = 1] - \mathbb{E}[Y \mid L = 0]$, and the lower bound tightens substantially. For Level 1 lobbying, MTR is less clearly justified (dimension-selection can have non-monotone effects on any specific policy parameter).

Monotone treatment selection (MTS). The typology implies that actors who lobby have weakly more favourable expected outcomes even absent lobbying, because they select into lobbying when conditions are favourable:

$$\mathbb{E}[Y(0) \mid L = 1] \;\geq\; \mathbb{E}[Y(0) \mid L = 0]. \tag{5}$$

Under MTS, the naïve difference is an even stricter upper bound on the true ATE, because part of the observed difference is due to selection rather than treatment.

Conditioning on the primitives. The tightest bounds come from conditioning on A (alignment) and P (power). Within strata defined by observable proxies for alignment and power, the confounding from these variables is absorbed (to the extent the proxies capture the true values). The overall ATE is a weighted average of the stratum-specific bounds:

$$\text{ATE} = \sum_{a,p} \text{ATE}(a,p) \cdot P(A = a, P = p), \tag{6}$$

where each $\text{ATE}(a,p) \in [L(a,p),\, U(a,p)]$ is bounded within the stratum. Without the typology, a researcher doing Manski bounds would condition on ad hoc covariates or not condition at all. With the typology, the conditioning set is principled: alignment (measurable through material-interest indicators) and power (measurable through economic weight and lobbying resources).

Level-specific bounds. The typology predicts that the effect of lobbying varies by level. For Level 3 lobbying on specific regulatory parameters, strong MTR applies (tight bounds). For Level 1 lobbying on agenda composition, MTR is weaker and the outcome space is less well-defined (wider bounds). Level-specific bounding acknowledges that the estimand is heterogeneous across levels.

A worked example. Consider the effect of incumbent lobbying on free-allocation shares in the EU Emissions Trading System. The outcome Y is the share of free allocation (bounded in $[0, 100\%]$). The treatment L is whether a sector association submitted comments to the allocation consultation. Alignment A is measurable (carbon intensity determines exposure). Power P is measurable (sector employment, lobbying expenditure).

1. *Worst-case bounds*: $[-100\%, +100\%]$. Uninformative.
2. *Impose MTR*: lobbying for free allocation cannot reduce free allocation. Upper bound: naïve difference. Lower bound: 0.
3. *Impose MTS*: firms that lobby are firms that would have received weakly more free allocation anyway (large, carbon-intensive, politically connected). Upper bound tightens below the naïve difference.

4. *Condition on alignment and power*: within strata of similar carbon intensity and firm size, compute conditional bounds. Remaining variation in lobbying is less confounded.
5. *Report*: bounds alongside the naïve point estimate. The gap between the naïve estimate and the MTS-tightened upper bound measures the portion of the observed association attributable to selection versus treatment.

4.6 Additional design strategies

Exogenous shocks to alignment. Instrument lobbying strategy with shocks that change alignment without directly affecting the policy outcome. Candidates: exogenous supply-chain shocks (global commodity prices, foreign technology costs, extreme weather events that change climate-risk exposure). The exclusion restriction—that the shock affects policy only through its effect on lobbying—is plausible for foreign shocks but less so for domestic ones. Limitation: this identifies the effect of *alignment-driven* lobbying but not the effect of lobbying holding alignment fixed. If lobbying is merely a conduit for alignment, the IV estimate captures the alignment effect, not the lobbying effect.

Discontinuity designs in the administrative arena. Regulatory thresholds (firms above a size threshold face different rules) provide quasi-experimental variation in lobbying incentive that is locally exogenous. Limitation: identifies only the local average treatment effect at the threshold, which may not generalise; and firms may sort around thresholds.

Difference-in-differences around the feedback operator. Policy events that shift alignment for some actors but not others (e.g. introduction of green industrial policy benefiting clean-energy firms but not fossil-fuel firms) create a natural experiment: compare the change in lobbying strategy and subsequent policy outcomes for affected versus unaffected actors. Limitation: requires the policy event to be plausibly exogenous to the actors being studied.

4.7 What honest empirical work looks like

The typology implies that credibly estimating lobbying effects is substantially harder than the existing literature acknowledges, for a specific reason: the same variables that determine lobbying strategy also determine policy outcomes through channels that do not run through lobbying. Any study that cannot separate the lobbying channel from the alignment and structural-power channels produces biased estimates. The direction of bias is predictable from the typology: upward for aligned actors, downward for misaligned actors, attenuated when the wrong level is measured, and inflated when observable lobbying proxies for unobservable relational influence.

This does not render the empirical literature worthless—but it does mean that findings must be interpreted *through* the bias structure. A finding that "pro-climate lobbying is associated with stronger climate policy" is more likely to reflect alignment confounding than lobbying effectiveness. A finding that "anti-climate lobbying has no measurable effect" is more likely to reflect Level 1 and relational-channel effects that the study could not measure than actual ineffectiveness.

The typology's greatest empirical value may not lie in telling us how to estimate lobbying effects but in telling us *why we cannot estimate them precisely* and in providing the structure for honest partial identification. A literature that reported Manski bounds conditioned on alignment and power—and that matched the level of measured lobbying to the level of the dependent variable—would be more epistemically honest, and potentially more useful, than the current literature of point estimates resting on untestable identifying assumptions.

There is an inverse relationship between the *magnitude* and *measurability* of lobbying effects. Level 1 strategies (dimension selection) are the most consequential but the hardest to observe and bound. Level 3 strategies (direct policy engagement) are the most measurable but the least consequential. The inferential frontier lies in the middle: Level 2 lobbying (constraint shaping) is moderately observable (through campaign expenditure, media analysis, agenda-tracking) and moderately consequential, making it the most promising target for credible estimation.

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